

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Even Semester)

Innovative Practice Description for Unit-I

Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC8691 Microprocessors and Microcontrollers

Name of the Faculty member: Mr.A.Azhagu Jaisudhan Pazhani

Name of the Topic: 8086 Microprocessor Architecture & Instruction Set of 8086, Assembler Directives and Interrupts and Interrupt Service Routines

Name of the Innovative Practices: Zero Minute Speech, One Minute Paper and Animation Video

Date and Duration: 18.12.2019 (10 Minutes), 06.01.2020 (10 Minutes) and 12.01.2020 (10 Minutes)

Description:

(i) Zero Minute Speech:

- The activity Zero Minute Speech is an impromptu speech where the speaker is supposed to express the idea(s) on the given topic, within a minute.
- The important topics were allotted to set of students on spot and they will give a speech on the topic with doubt clarification among classmates.

(ii) One Minute Paper:

This method used at the end of class for assessing student performance. Advantage of this technique is that it provides quick feedback on whether the concept is reached to the students. Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. Sometimes, instead of asking for the main point, a professor may wish to probe for the most disturbing or most surprising item.

(iii) Animation video

- Use of audio-visual aids help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

Goals (Learning Outcomes):

- The students will be able to describe the architecture of microprocessor 8086.
- The students will be able to gain knowledge on assembler directives.
- The students will be able to explain about interrupts and interrupt service routines.

Use of appropriate method:

Justification for choosing the following Activities:

- **Zero Minute Speech:** Students can able to express basic concepts of particular topics.
- **Minute Paper:** Advantage of minute papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.
- **Animation Video:** Animation provides opportunities for real communication between teachers and students in learning.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- **Zero Minute Speech:** All the students are participated in this activity.
- **Minute Paper:** Students used small paper to write a few points about assembler directives with examples.
- **Animation:** Students can easily understand the interrupt service routine of 8086 microprocessor and also give some input to recall the information from their conventional learning.

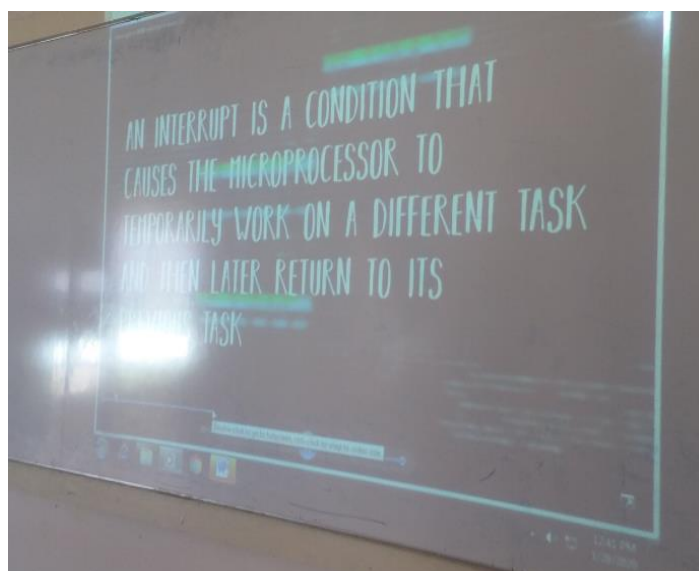
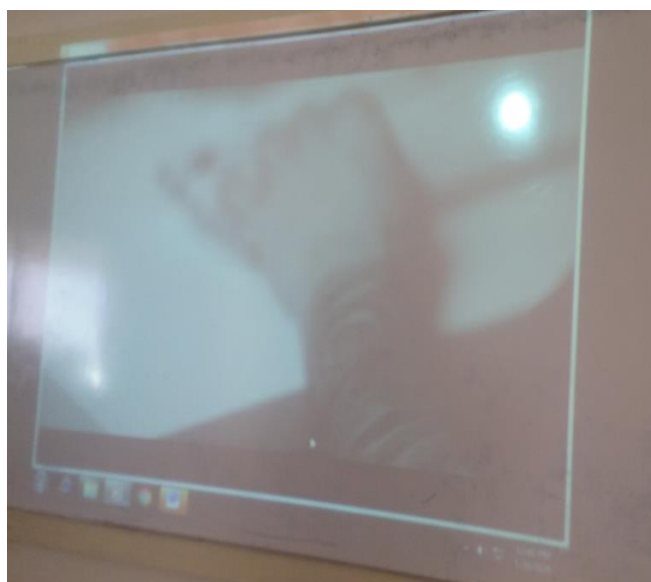
**Zero Minute Speech for 8086 Microprocessor Architecture & Instruction Set
on 18.12.2019 (10 Minutes)**



One Minute Paper for Assembler Directives on 06.01.2020 (10 Minutes)



**Animation Video for Interrupts and Interrupt Service Routines
on 12.01.2020 (10 Minutes)**



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Zero Minute Speech:** With interest, most of the students delivered the topics with examples
- **Minute Paper:** Helps the students to recall the important assembler directives and their functions with examples.
- **Animation:** Interrupt service routine explained in step by step through animation.

Changes required for future (if required):

- Students requested to give topics before a day for their preparation for zero minute speech.
- Sometime need to explain a few concepts during animation video presentation with real-time example so more time has to allocate for this activity in the future.

Reflective Critique:

Challenges:

- **Zero Minute Speech:** During zero minute speech few students hesitated to participate in the activity.
- **Minute Paper:** Students struggled to recollect the points
- **Animation:** Need to give real-time examples for their better understanding of the topic.

Benefits:

- **Zero Minute Speech:**
 - Able to recall few topics quickly and students can able to link the topic with real-time examples.
- **Animation:**
 - The animation is eye-catching, It is useful when quickly getting and holding the audience's attention.
 - The animation can show the imagined objects in the gesticulation, It is ideal for demonstrating processes.
 - Interactive learning with live-action animation keeps learners interested and reinforces skills.
- **Minute Paper:**
 - Provide a “conceptual bridge” between successive class periods.
 - Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	1	-	2	-	1	-	1

CO1: The students will be able to describe the architecture of microprocessor 8086 and execute programs based on 8086 microprocessor.

References:

- ❖ <http://www.icaltefl.com/just-a-minute-speaking-activity>
- ❖ <https://busyteacher.org/20564-5-steps-to-making-fluency-fun-just-a-minute.html>
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- ❖ <https://www.edutopia.org/blog/design-thinking-one-minute-papers-ashley-nahornick>
- ❖ <https://www.youtube.com/watch?v=NXI7NTE95QY>
- ❖ Yu-Cheng Liu, Glenn A.Gibson, “Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design”, Second Edition, Prentice Hall of India, 2007.